



VLSI Testing

積體電路測試

Memory Testing

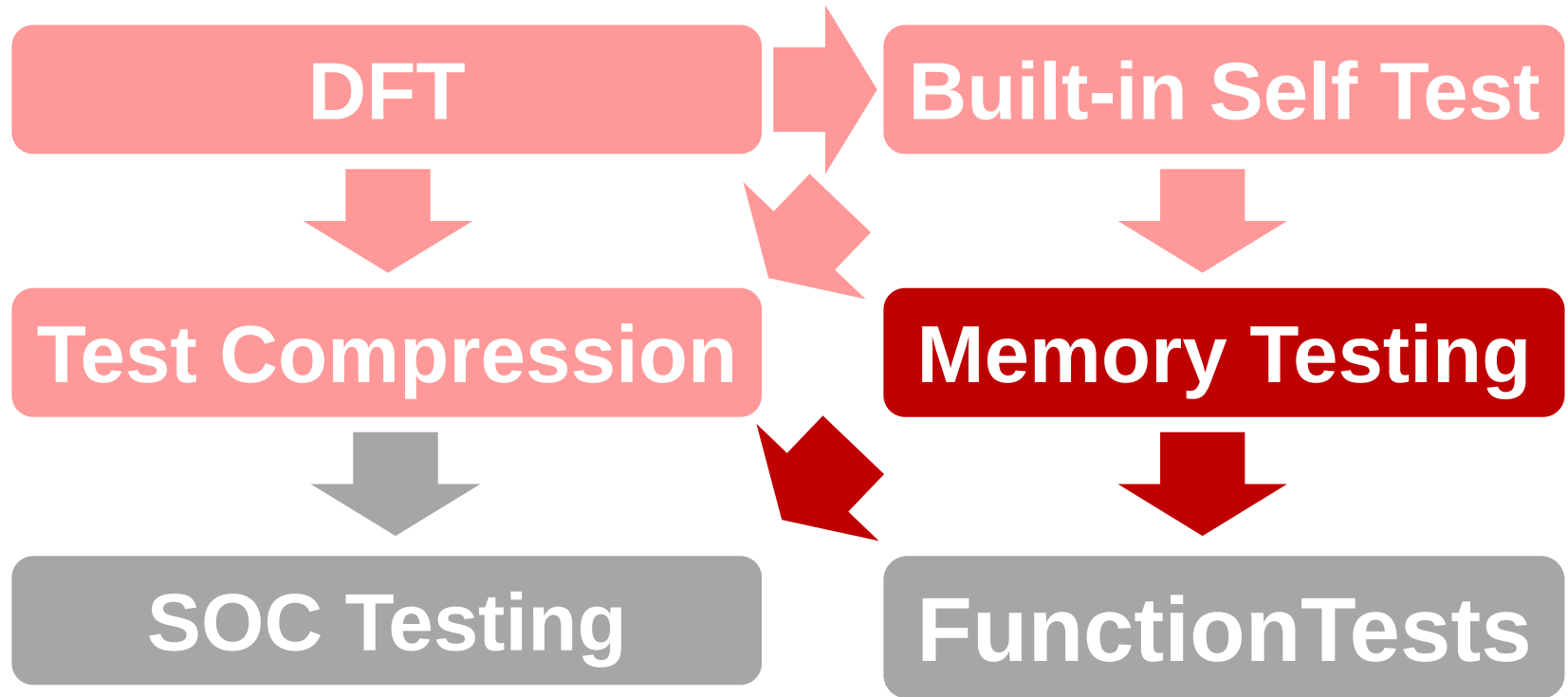
Professor James Chien-Mo Li 李建模

Lab. of Dependable Systems

Graduate Institute of Electronics Engineering

National Taiwan University

Course Roadmap (Design Topics)



Why Am I Learning This?

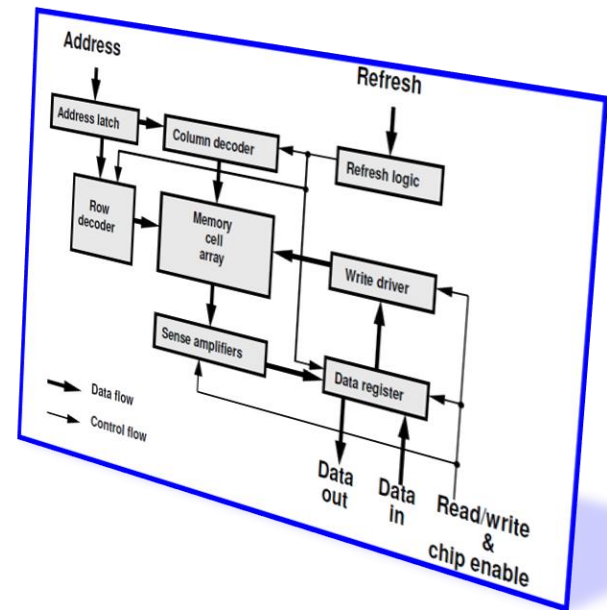
- Testing memory is important
 - ♦ Because many memories are needed in modern designs
- Memory testing is very different from logic testing

“The advantage of a bad memory is that one enjoys several times the same good things for the first time”

(Friedrich Nietzsche)

Outline

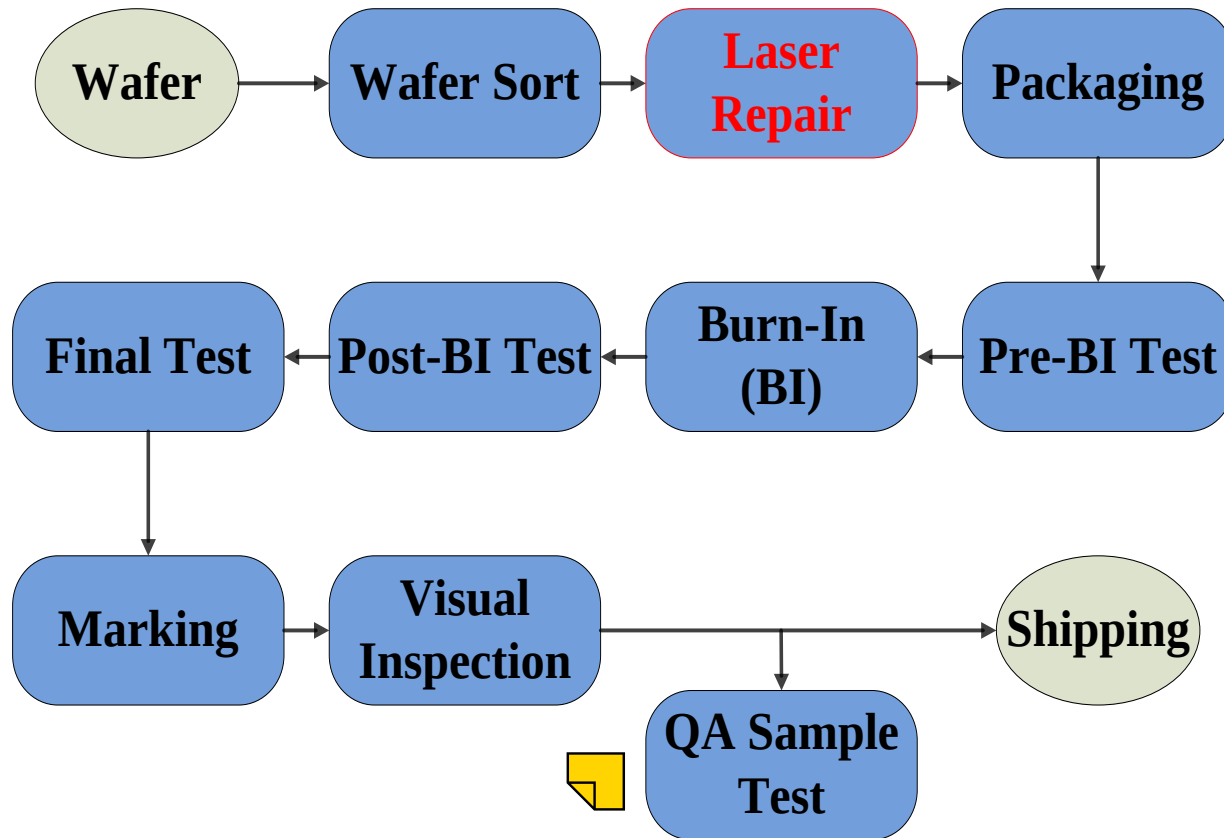
- Introduction
- Memory Fault Models
- Memory Test Algorithms
- Memory Fault Simulation (*not in exam)
- Memory Test Generation (*not in exam)
- Memory BIST (*not in exam)



Memory Testing

- Testing of memory is crucial for quality. Research started 1960's
 - ♦ Memory accounts for ~30% of semiconductor market (2019)
- Popular memory test items
 - ♦ Off-chip tests
 - * DC parametric test: e.g. leakage current, output voltage level
 - * AC parametric test: e.g. rise time, fall time
 - * Functional test: e.g. march test (see 16.3)
 - * Retention test: measure retention time of DRAM
 - * Reliability test (Burn-in)
 - ♦ On-chip tests
 - * Error Detection and Correction (EDAC): on-line testing
 - * Built-in Self Test (BIST)
 - * Built-in Self Diagnosis & Repair (BISDR)
- BIST important for System-on-chip (SOC) with embedded memories

Typical Memory Test Flow

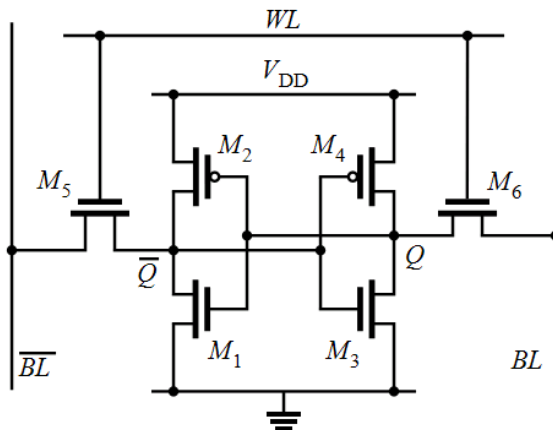


Memory Diagnosis/Repair can Improve Yield

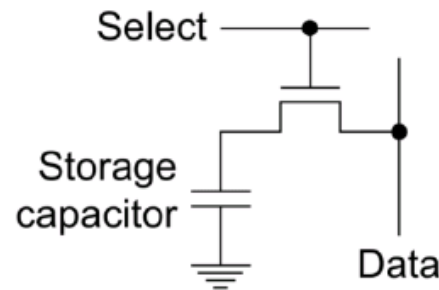
Types of Memories



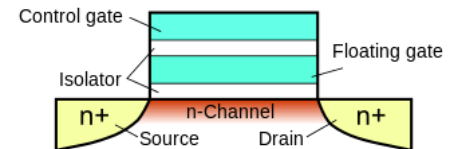
type	area	speed	retention	application	test method
SRAM	largest 6T	fastest <1ns	as long as power	embedded SRAM cache, registers	BIST
DRAM	medium 1T+1C	medium ~10ns	< sec.	embedded DRAM	BIST/ ATE
				on-board memory	ATE
Flash	smallest 1T	slowest ~100μs	years	SSD, USB drive	ATE



SRAM cell [wikipedia]

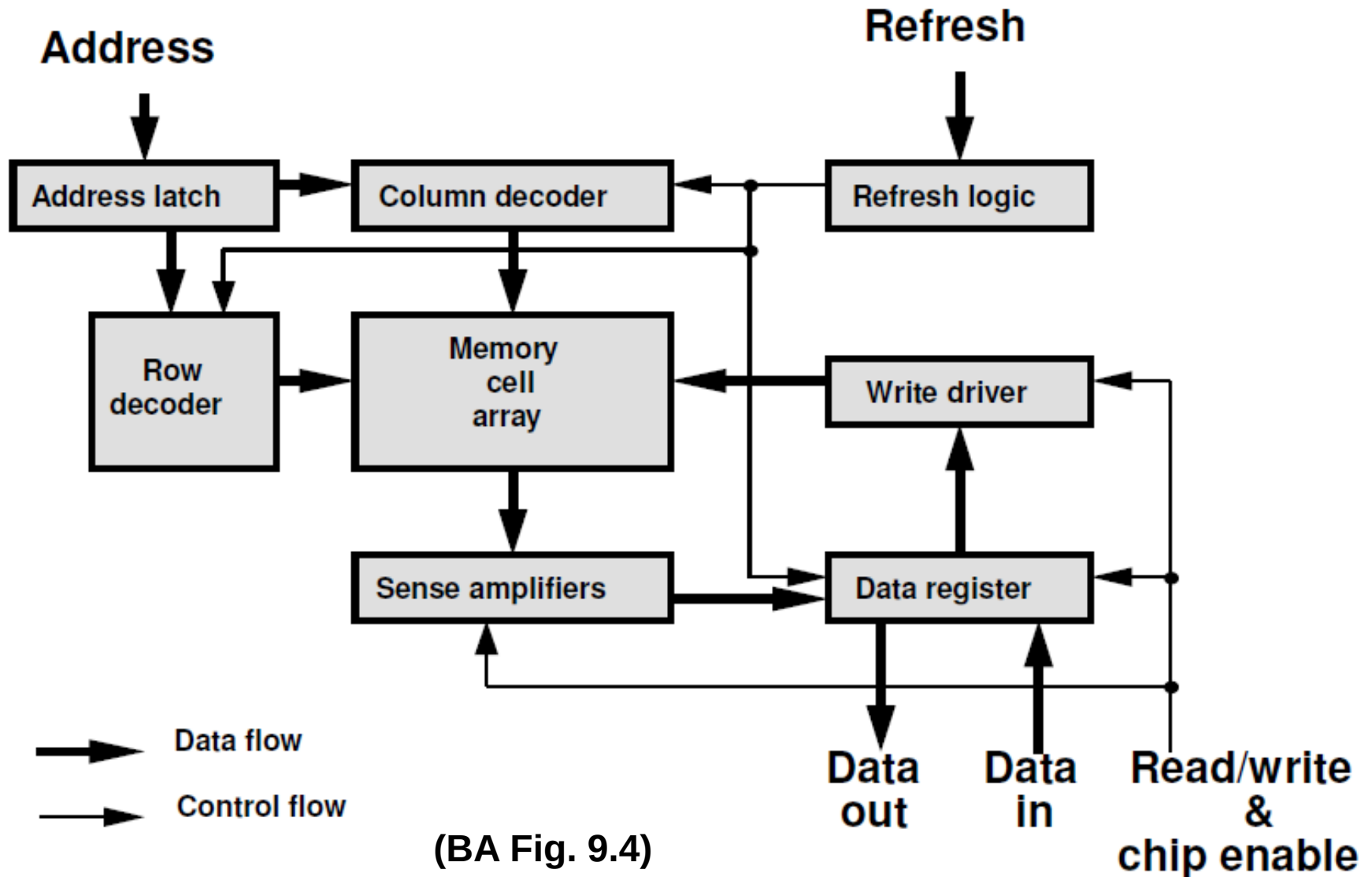


DRAM cell



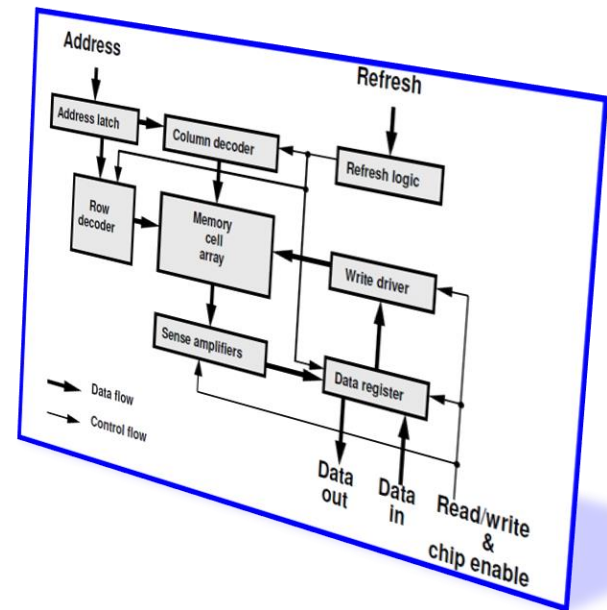
Floating Gate Transistor
[wikipedia]

DRAM Functional Model



Outline

- Introduction
- Memory Fault Models (focus on RAM)
 - ◆ Static faults
 - * Single cell fault, Double cell fault, Address-decoder fault
 - ◆ Dynamic faults
 - * Recovery fault, Retention fault
- Memory Test Algorithms
- Memory Fault Simulation (*not in exam)
- Memory Test Generation (*not in exam)
- Memory BIST (*not in exam)



RAM Fault Models

- **Functional testing** is commonly used for memories
 - ♦ Scan testing for logic is not applicable to memory. (Why? FFT)
- **Functional fault models** are behavior model for faulty memories
 - ♦ based on **real defects**, not imagination
- Popular **RAM functional fault models**
 1. **Static fault models**: faulty behavior does NOT change with time
 - * single cell, double cell, address decoder
 2. **Dynamic fault models**: faulty behavior changes with time
 - * recovery, data retention
- NOTE: many other RAM fault models (neighborhood pattern sensitive faults ...)

Different Memories Need Different Fault Models

RAM Fault Models (1) – single cell

- **Stuck-At Fault (SAF)**
 - ♦ a cell is always 0, **SA0**
 - ♦ a cell is always 1, **SA1**
- **Stuck-Open Fault (SOF)**
 - ♦ a cell cannot be accessed due to broken wire
- **Transition Fault (TF)**
 - ♦ a cell fails to
 - * Rise from 0 to 1 **<↑ / 0>**
 - * Fall from 1 to 0 **<↓ / 1>**

NOTATION: **<S/F>**: a fault in a cell [van de Goor 91]

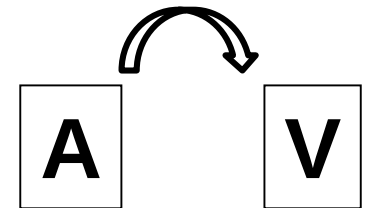
- ♦ **S** is value or operation activating fault , $S \in \{0, 1, \uparrow, \downarrow, \updownarrow, \forall\}$
 - * **↑** is rising; **↓** is falling; **↑↓** is either **↑** or **↓**; **∀** means any condition
- ♦ **F** is faulty value of cell , $F \in \{0, 1, \updownarrow\}$
 - * **↑↓** is complement

RAM Fault Models (2) – double cell

- **Coupling Fault (CF):** **Victim cell** is affected by **Aggressor cell**
 - ◆ **1.State Coupling Fault (CF_{st})**
 - * if aggressor cell is in given state, victim cell is forced to 0 or 1
 - * 4 types: $\langle 0; 0/1 \rangle$ or $\langle 0; 1/0 \rangle$ or $\langle 1; 1/0 \rangle$ or $\langle 1; 0/1 \rangle$
 - ◆ **2.Inversion Coupling Fault (CF_{in})**
 - * if aggressor cell rise/fall, victim cell is complemented
 - * 2 types: $\langle \uparrow; \nabla/\downarrow \rangle$ or $\langle \downarrow; \nabla/\uparrow \rangle$
 - ◆ **3.Idempotent Coupling Fault (CF_{id})**
 - * if aggressor cell rise/fall, victim cell is forced to 0 or 1
 - * 4 types: $\langle \uparrow; 0/1 \rangle$ or $\langle \uparrow; 1/0 \rangle$ or $\langle \downarrow; 0/1 \rangle$ or $\langle \downarrow; 1/0 \rangle$

NOTATION: $\langle S_1; S_2/F \rangle$ faults in 2 cells

- ◆ S_1 is value or operation activating fault in aggressor
- ◆ S_2 is value or operation activating fault in victim
- ◆ F is faulty value of victim



CF Examples

$CF_{st} < 0, 0/1 >$

addr	content*
A	0
V	0

↓ do nothing

addr	content
A	0
V	0/1

↓ read V

detected

$CF_{id} < \uparrow; 0/1 >$

A	0
V	0

↓ write 1 to A

A	1
V	0/1

↓ read V

detected

A	0
V	1

↓ write 1 to A

A	1
V	1

↓ read V

not detected

$CF_{in} < \uparrow; \forall I \updownarrow >$

A	0
V	0

↓ write 1 to A

A	1
V	0/1

↓ read V

detected

A	0
V	1

↓ write 1 to A

A	1
V	1/0

↓ read V

detected

*typically many bits in one address
but only one bit here for illustration

QUIZ

Q: Consider $CF_{in} \langle \downarrow, \uparrow \rangle$. Fill in values for two cases.
Can we detect faults in both cases?

ANS:

ad dr	content
A	1
V	1

↓ write 0
to A

ad dr	content
A	0
V	?

↓ read
V=?

ad dr	content
V	1
A	1

↓ write 0
to V

ad dr	content
V	0
A	?

↓ read
A=?

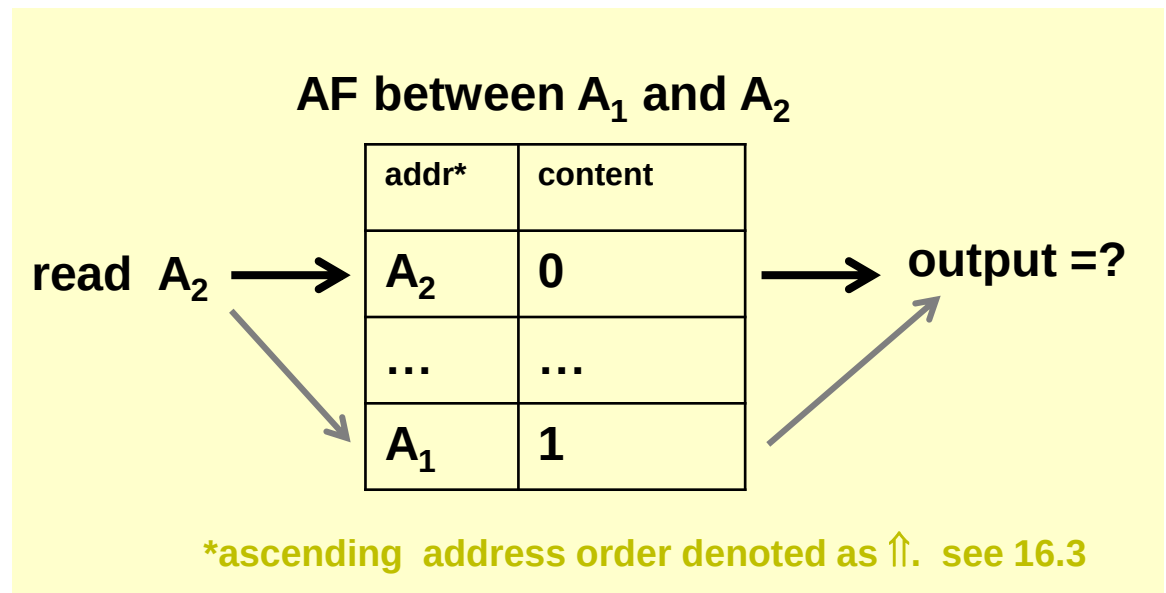
Address Order Matters

RAM Fault Models (3) – AF

- **Address-Decoder Fault (AF)** Four faulty behavior:
 1. Given a certain address, no cell will be accessed
 2. A certain cell is never accessed by any address
 3. A certain cell can be accessed by multiple addresses
 - ★ 4. Given a certain address, multiple cells are accessed

➤ **AND-type**

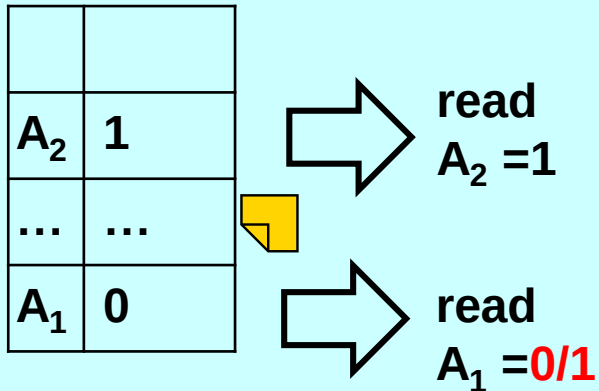
➤ **OR-type**



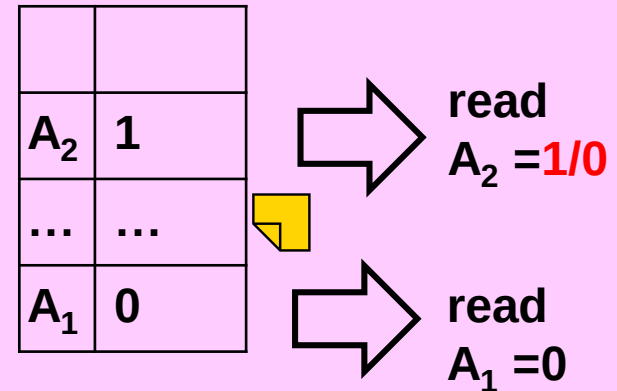
Only Consider #4 AF

AF Examples

OR-type AF
between A_1 and A_2

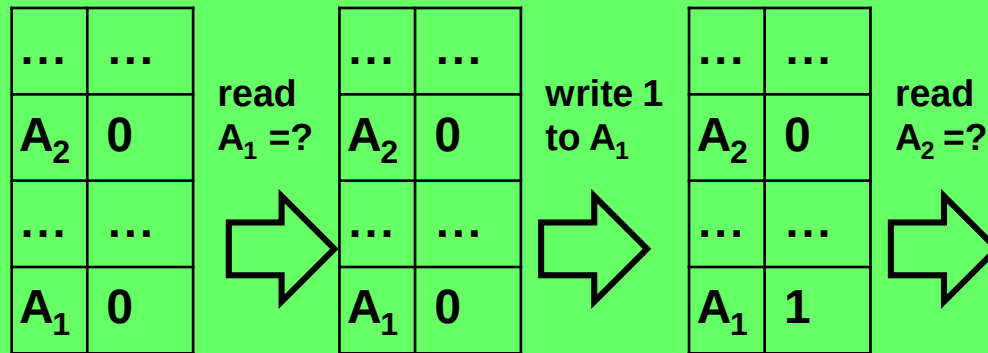


AND-type AF
between A_1 and A_2

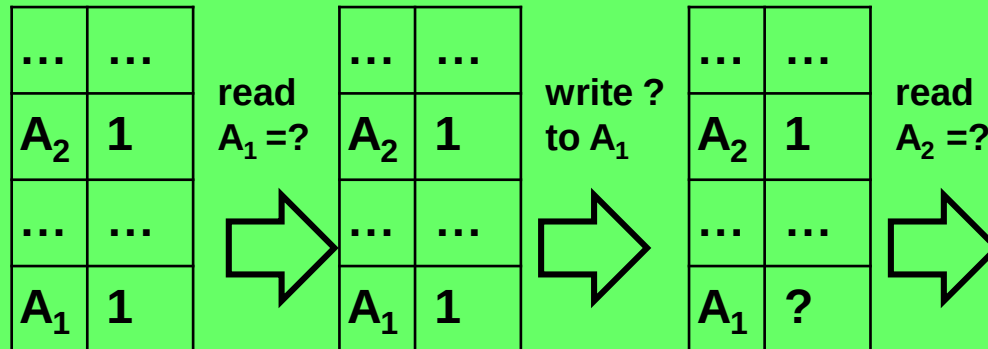


QUIZ

Q1: Given **OR-type** AF between A_1 and A_2 . Fault detected?



Q2: Given **AND-type** AF. Find a test to detect fault.



AND/OR AF Need Opposite Test Data

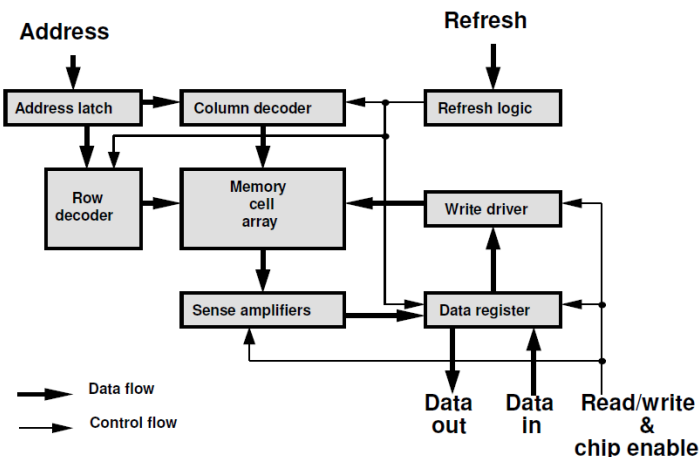
RAM Fault Models (4) – dynamic faults

- **Data Retention Fault (DRF)**: data changed after a certain time
 - ♦ **DRAM** 
 1. Charge leakage loose data in capacitor
 2. Refresh logic fails to refresh correctly
 - ♦ **SRAM**
 - * Defective pull-up device inducing excessive leakage current which changes the state of cell
- **Sense amplifier recovery fault** 
 - ♦ Sense amp. saturated after reading/writing a long string of 0 or 1
- **Write recovery fault** 
 - ♦ A write followed by a read/write at a different location results in reading or writing at the same location due to slow address decoder

Time Consuming to Test Dynamic Faults

Summary

- Memory test important. Good diagnosis/repair can **improve yield**
- Popular **RAM functional fault models**
 - ◆ **Static** fault models
 - * Single cell: Stuck-at (**SAF**), Stuck-open (**SOF**), Transition (**TF**)
 - * Double cell coupling faults: **CF_{in}**, **CF_{id}**, **CF_{st}**
 - * Address decoder fault (**AF**): **AND**-type **OR**-type
 - ◆ **Dynamic** fault models
 - * Data retention faults (**DRF**). Recovery faults
- Fault model must be **realistic**
 - ◆ Different memories need **different** fault models



FFT

- Q1: Scan testing for logic is not applicable to memory.
 - ◆ Why no scan? 
- Q2: AF has four faulty behavior.
 - ◆ We only consider #4. The others are easy to test, why?
 1. Given a certain address, no cell will be accessed
 2. A certain cell is never accessed by any address
 3. A certain cell can be accessed by multiple addresses
 - ★4. Given a certain address, multiple cells are accessed